

Question 1 Which of the following is logically equivalent to the given statement: *If I like Dunkin' Donuts, then I like donuts.*

If I like donuts, then I like Dunkin Donuts.

If I do not like Dunkin' Donuts, then I do not like donuts.

If I do not like donuts, then I do not like Dunkin' Donuts. ✓

I like Dunkin Donuts, then I do not like donuts.

Question 2 Which of the following is logically equivalent to the given statement: *If my plant is an African violet, then its flowers are purple.*

If the flowers on my plant are purple, then my plant is an African violet.

If my plant is not an African violet, then its flowers are not purple.

If the flowers on my plant are not purple, then the plant is not an African violet.
✓

The flowers on my plant are purple if and only if the plant is an African violet.

Question 3 Label each statement as either the **contrapositive**, **converse** or **inverse** of the given statement. If it is none of them, write **NA**.

"If I am a CS major, I will take Discrete Math".

- If I am not a CS major, I will take Discrete Math.
 - If I am a CS major, I will not take Discrete Math.
 - If I dont take Discrete Math, I am not a CS major.
 - If I am not a CS major, I will not take Discrete Math.
 - If I take Discrete Math, then I am a CS major.
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Question 4 Rewrite the following as either $p \rightarrow q$ or $q \rightarrow p$. These questions might be difficult but are simply meant for practice, so just give them your best shot! Try replacing p and q in either order with simple expressions where the implication works only one way (e.g., x is a cat, x is an animal) and see which way sounds correct. **Enter 1 for $p \rightarrow q$ and 2 for $q \rightarrow p$**

- p implies q :
- p if q :
- p only if q :
- p is necessary for q :
- p is sufficient for q :
- p because q ($p \therefore q$):
- p therefore q ($p \therefore q$):
- $\neg p$ unless q :

Question 5 Fill in the truth table for $p \leftrightarrow q$, using T and F wherever required. Which logical equivalences does this table prove?

p	q	$\sim p$	$\sim q$	$p \rightarrow q$	$\sim q \rightarrow \sim p$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$	$p \leftrightarrow q$
T	T	F	F	T	T	T	T	T
T	F	F	T	F	F	T	F	F
F	T	T	F	T	T	F	F	F
F	F	T	T	T	T	T	T	T

Question 6 Syllogisms. Choose the statement that connects all statements above and is true.

- Cute animals are being unconditionally loved.
- No cat is incapable of purring.
- Animals being unconditionally loved can one day rule the world.
- No animal is not cute who can purr.

Multiple Choice:

- (a) Animals that will rule the world are cute.
- (b) Animals that are being unconditionally loved are cats.
- (c) Cats will rule the world. ✓
- (d) If an animal is being unconditionally loved, then it can purr.

Question 7 *Syllogisms. Choose the statement that connects all statements above and is true.*

- *No one who lives on Beacon Street doesn't pursue a BFA degree.*
- *Those who have taken a course at the SMFA love art.*
- *No one who is pursuing a BFA fails to take a course at the SMFA.*

Multiple Choice:

- (a) *If someone is a BFA student, then they live on Beacon Street.*
 - (b) *Those who love art pursue BFA degrees.*
 - (c) *If someone lives on Beacon Street, then they love art. ✓*
 - (d) *Those who have taken a course at the SMFA live on Beacon Street.*
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